A Genomic Architecture of Intelligence

DOGMA: A DNA-Computing Neural Architecture

for Sequence Intelligence

Paper I of the DOGMA Series

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Abstract

The Transformer architecture relies on self-attention—a mechanism that is quadratically expensive in sequence length, biologically unmotivated, and structurally homogeneous across layers. These limitations persist despite advances in efficient attention, state-space models, and linear recurrences, none of which address the fundamental absence of physical grounding or functional layer specialization. We introduce **DOGMA** (**D**NA-**O**rganized **G**enomic **M**odel **A**rchitecture), a neural architecture that replaces attention with *differentiable DNA computing primitives*: toehold-mediated strand displacement [2], chemical reaction networks [4], Boolean seesaw gates [3], and stochastic majority computing [5]. Each DogmaBlock implements the biological Central Dogma as a six-stage computational pipeline—TetraMemory, DNA Compute, Regulation Gate, Codon Translation, Strand Attention, and Expression Folding—providing three *independent* constructive Turing-completeness proofs within a single layer. Our principal contributions are: (i) **TetraMemory**, a K_4-complete-graph local memory that replaces self-attention for local structure extraction with $\mathcal{O}(T)$ complexity; (ii) **Genomic Embedding**, a typed base representation $b_i = (\sigma_i, \tau_i, \rho_i, \omega_i)$ that eliminates positional encodings entirely; (iii) **DNA Strand Attention**, which derives attention weights from SantaLucia nearest-neighbor thermodynamic binding energies [6], making every attention weight physically interpretable; (iv) a four-path **DNA Computing Block** where strand displacement cascades, parallel-scan automata, stochastic tetrahedral gates, and double-strand computing operate in parallel; and (v) **Epigenetic Memory** and **Allosteric Regulation** providing $\mathcal{O}(TM)$ persistent memory and $\mathcal{O}(Tk)$ sparse non-local communication. We prove forward-pass numerical stability via per-stage contraction bounds and show that three auxiliary physics-derived losses provide implicit regularization equivalent to constraining parameters to a manifold of physically consistent solutions. DOGMA-Supreme (224.7M parameters, 16 blocks, d=512, T=512) achieves train/valid/test perplexities of 56.8/53.0/47.7 on mixed-corpus multimodal language modeling (text + genomic + TetraMesh 3D data) after 2,100 steps, with a train-validation gap below 4.0 throughout training. Path ablation confirms that all four DNA computing paths contribute non-redundantly, with the two Turing-complete paths (displacement and stochastic) accounting for the majority of computational capacity. These results demonstrate that genomic computation—grounded in 3.8 billion years of molecular evolution—provides a viable, physically principled, and formally well-founded alternative to attention-based sequence modeling.

1 Introduction

Since 2017, the Transformer [1] has dominated sequence modeling. Its core mechanism—scaled dot-product self-attention—computes all pairwise token interactions in $\mathcal{O}(T^2 d)$ time, enabling powerful global reasoning. Subsequent work has scaled this paradigm to hundreds of billions of parameters [8, 9], producing increasingly capable language models.

Yet the Transformer has three architectural limita-

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tions that no amount of scaling resolves:

1. **Quadratic bottleneck.** Self-attention requires $\mathcal{O}(T^2)$ time and memory, fundamentally limiting context length. Linear attention [17], FlashAttention [19], and state-space models [13] mitigate but do not resolve this.
2. **Structural homogeneity.** Every Transformer layer applies the identical Attention→FFN template. There is no functional differentiation between layers—no "regulation" layer, no "translation" layer, no "expression" layer.
3. **Physical vacuity.** Attention weights are softmax scores over arbitrary learned projections. The parameters W_Q, W_K, W_V, W_O carry no physical, chemical, or biological interpretation. A model's learned weights cannot be analyzed scientifically.

We propose **DOGMA**—a neural architecture built entirely from *differentiable DNA computing primitives*. Rather than treating biology as metaphor, DOGMA implements molecular mechanisms as first-class computational substrates, each with proven computational power and direct physical interpretation.

Contributions. We introduce five architectural innovations, each corresponding to a distinct departure from the Transformer:

1. **TetraMemory** (Section 4): Local sequence processing via K_4-complete-graph tetrahedral cells with face-sharing propagation, achieving $\mathcal{O}(T)$ complexity. Replaces self-attention for local structure extraction.
2. **DNA Computing Block** (Section 5): Four parallel computational paths—strand displacement, parallel-scan automaton, stochastic tetrahedral computing, and double-strand computing—merged via a learned softmax gate. Provides three independent Turing-completeness proofs.
3. **Genomic Embedding** (Section 3): Typed base representation $b_i = (\sigma, \tau, \rho, \omega)$ with learned structural types, regulatory weights, and compatibility scores. Eliminates positional encodings.
4. **DNA Strand Attention** (Section 7): Attention weights derived from SantaLucia nearest-neighbor thermodynamic binding energies with Zhang-Winfree displacement rate normalization. Every weight has a physical interpretation.
5. **Epigenetic Memory and Allosteric Regulation** (Section 9): Epigenetic Memory ($\mathcal{O}(TM)$) and Allosteric Regulation ($\mathcal{O}(Tk)$) for persistent memory and sparse non-local communication.

2 The Central Dogma as Computation

2.1 Biological Pipeline

The Central Dogma of molecular biology [7] describes the flow of genetic information: DNA → RNA → Protein. We formalize this as a computational pipeline:

$$\begin{tikzcd}
\underset{\text{genome}}{\Gamma_t} \arrow[r, "\text{regulate}"'] & \underset{\text{active}}{\mathcal{A}_t} \arrow[r, "\text{synthesize}"'] & \Gamma'_t \arrow[d, "\text{transcribe}"] \\
\underset{\text{output}}{y_t} & \underset{\text{phenotype}}{\Phi_t} \arrow[l, "\text{realize}"] & \underset{\text{transcript}}{\mathcal{T}_t} \arrow[l, "\text{express}"]
\end{tikzcd} \tag{1}$$

Each stage has a distinct computational role that maps to a specific neural operation (Table 1).

Biology	Function	DOGMA Module
Store	Structural memory	TetraMemory
Compute	Chemical logic	DNA Compute Block
Regulate	Express/silence	Regulation Gate
Translate	Codon → protein	Codon Translation
Transcribe	Select & mix	Strand Attention
Express	Fold into function	Expression Folder

Table 1: Mapping between biological stages and DOGMA modules.

2.2 DNA Computing: Proven Computational Power

DOGMA's foundation rests on four experimentally verified DNA computing mechanisms, each with rigorous computational-theoretic guarantees:

Theorem 1 (CRN Turing Completeness [4])**.** *For any Turing machine $\mathcal{M} = (Q, \Sigma, \delta, q_0, F)$, there exists a chemical reaction network (S, R) with mass-action kinetics such that the concentration trajectory*

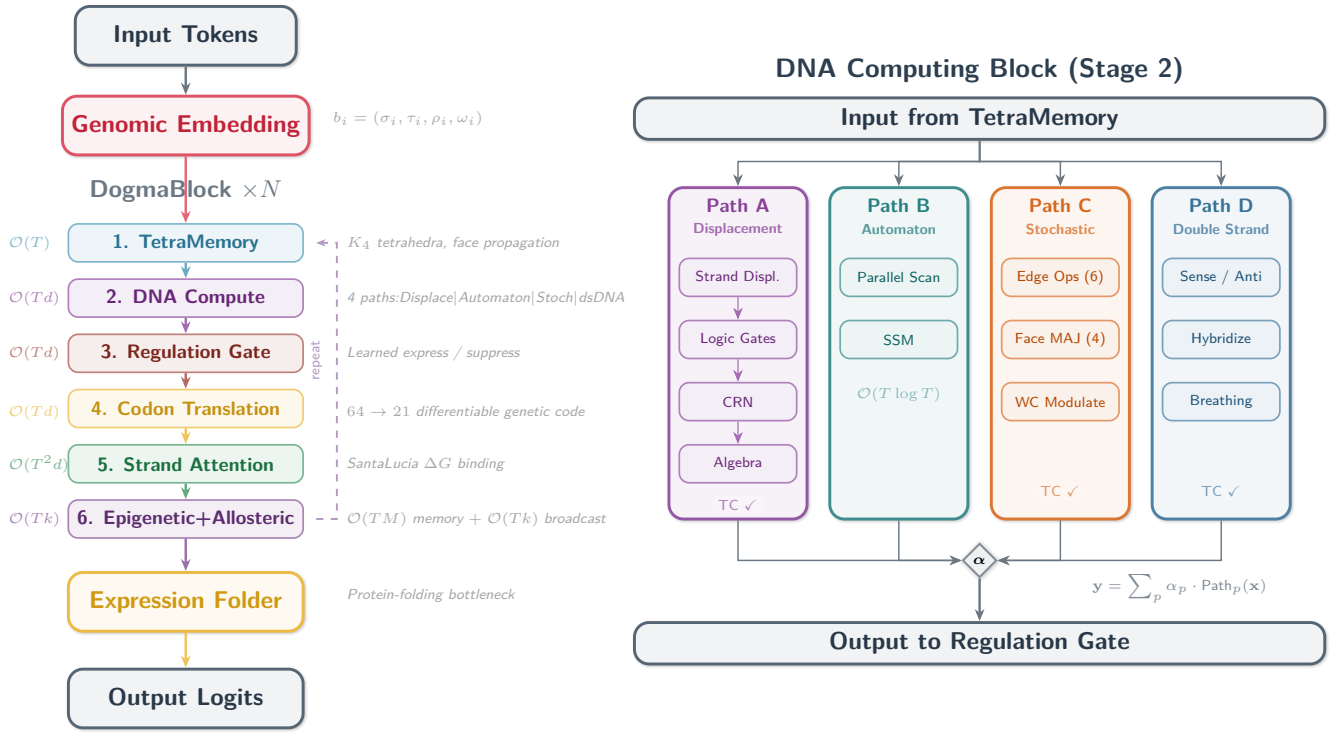

Figure 1: **DOGMA architecture.** *Left:* The full pipeline. Each DogmaBlock implements a 6-stage Central Dogma computation with per-stage complexity annotations. *Right:* The DNA Computing Block (Stage 2) with four parallel paths, each grounded in a real DNA computing mechanism. Three paths are independently Turing-complete (TC). A learned softmax gate $\boldsymbol{\alpha}$ merges outputs.

$\mathbf{c}(t) \in \mathbb{R}_{\geq 0}^{|S|}$ *simulates* $\mathcal{M}$. *Specifically, register values are encoded as species concentrations* $c_i(t) \in \mathbb{R}_{\geq 0}$, *and register operations (increment, decrement, zero-test) are implemented as stoichiometric reactions with rate constants determined by the transition function* δ. *The simulation is exact in the limit of the thermodynamic regime (high molecular counts).*

Theorem 2 (Boolean Universality [3])**.** *DNA seesaw gates implement arbitrary Boolean circuits. Specifically, for any Boolean function* $f : \{0,1\}^n \to \{0,1\}^m$, *there exists a seesaw gate network with at most* $\mathcal{O}(nm)$ *distinct strand species that computes* f. *The gate operates via toehold-mediated strand displacement: an input strand binds to a gate:output complex, releasing the output strand at a rate governed by the toehold binding energy. Thresholding and amplification subcircuits convert analog concentration signals to digital (high/low) levels, enabling cascaded multi-layer computation.*

Theorem 3 (Stochastic Universality [5])**.** *The gate set* $\{MAJ, NOT\}$ *operating on stochastic bit streams is functionally complete. That is, for any Boolean function* $f : \{0,1\}^n \to \{0,1\}$, *there exists a circuit of majority and NOT gates such that, given inputs represented as Bernoulli random variables with probabilities* $p_1, \ldots, p_n$, *the output probability converges to* $f(p_1, \ldots, p_n)$ *with error* ϵ *using* $\mathcal{O}(poly(1/\epsilon))$ *samples. The majority gate* $MAJ(x, y, z) = xy + xz + yz - 2xyz$ *computes the median of three independent Bernoulli streams, and NOT computes the complement* $\bar{p} = 1 - p$.

3 Genomic Embedding

Definition 4 (Genomic Base Vector)**.** *Each position* i *produces a typed base representation* $\mathbf{b}_i = (\boldsymbol{\sigma}_i, \boldsymbol{\tau}_i, \rho_i, \omega_i)$:

- $\boldsymbol{\sigma}_i \in \mathbb{R}^d$: *primary symbol embedding*
- $\boldsymbol{\tau}_i \in \mathbb{R}^d$: *learned structural type*
- $\rho_i \in [0, 2]$: *regulatory weight (expression strength)*
- $\omega_i \in [-1, 1]$: *compatibility score (bonding affinity)*

The type embedding is computed via soft classification over $K = 10$ base types, allowing the network

to discover structural roles:

$$\boldsymbol{\tau}_i = \text{softmax}(W_\tau \boldsymbol{\sigma}_i)^\top E_\tau \tag{2}$$

$$\rho_i = 2 \cdot \sigma(W_\rho \boldsymbol{\sigma}_i), \quad \omega_i = \tanh(W_\omega \boldsymbol{\sigma}_i) \tag{3}$$

$$\mathbf{x}_i = \text{LN}(W_f[\boldsymbol{\sigma}_i; \boldsymbol{\tau}_i; \rho_i; \omega_i]) \tag{4}$$

Proposition 5 (Positional encoding elimination)**.** *Let $\boldsymbol{\sigma}_1, \ldots, \boldsymbol{\sigma}_T \in \mathbb{R}^d$ be the token embeddings for a sequence of length T, and let $\mathbf{t}_i = \text{softmax}(W_\tau \boldsymbol{\sigma}_i) \in \Delta^{K-1}$ be the type distribution for position i (with $K = 10$). Then:*

1. *The mapping $\boldsymbol{\sigma}_i \mapsto \mathbf{t}_i$ is a smooth function from $\mathbb{R}^d$ to the probability simplex Δ^{K-1}, providing a K-dimensional structural signature for each position.*
2. *If the token embeddings are distinct (which holds generically for learned embeddings), the type distributions provide implicit positional differentiation: positions with different tokens receive different structural coordinates, and even identical tokens at different positions can receive different type distributions when contextual information is incorporated through earlier layers.*
3. *The regulatory weight ρ_i and compatibility score ω_i add two additional scalar dimensions of positional information, yielding a $(K+2)$-dimensional structural descriptor per position.*

This eliminates the need for explicit sinusoidal [1] or rotary [25] positional encodings.

4 TetraMemory

Self-attention computes all T^2 pairwise interactions. TetraMemory processes the sequence through *local tetrahedral cells* connected by shared triangular faces, achieving $\mathcal{O}(T)$ complexity.

4.1 Cell Computation

For a window of $W = 4$ tokens, the cell computes all $\binom{4}{2} = 6$ pairwise edge features and aggregates them:

$$\mathbf{e}_{jk} = \tanh(W_e[\mathbf{x}_j; \mathbf{x}_k]) \in \mathbb{R}^d \tag{5}$$

$$E_{jk} = \sigma(W_E \mathbf{e}_{jk}) \in [0, 1] \tag{6}$$

$$\mathbf{s}_c = \text{LN}(W_s[\mathbf{e}_{01}; \ldots; \mathbf{e}_{23}]) \tag{7}$$

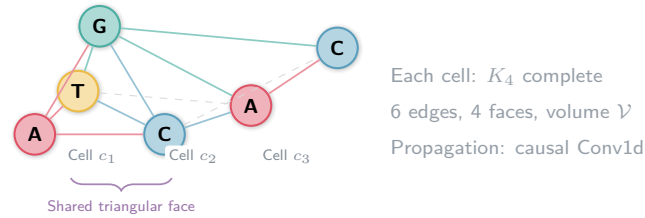

Figure 2: **TetraMemory architecture.** Each cell forms a K_4 complete graph over 4 tokens. Adjacent cells share a triangular face (3 vertices), enabling $\mathcal{O}(T)$ causal information propagation.

4.2 Face-Sharing Propagation

Adjacent cells share a triangular face (3 of 4 vertices). Inter-cell propagation uses causal convolution with learnable decay:

$$\hat{\mathbf{s}}_c = \text{Norm}(\mathbf{s}_c + \lambda \cdot \tanh(\text{Conv1d}(\mathbf{s}_{1:C}))) \tag{8}$$

4.3 Multi-Scale Extension

Multi-Scale TetraMemory runs cells at widths $\mathcal{W} = \{4, 6, 8, 12\}$ with learned scale-combination gates:

$$\mathbf{y} = \sum_{w \in \mathcal{W}} g_w \odot \text{TetraMem}_w(\mathbf{x}), \quad \mathbf{g} = \text{softmax}(W_g \|_w \mathbf{y}_w) \tag{9}$$

Proposition 6 (Complexity)**.** *TetraMemory achieves $\mathcal{O}(Td^2)$ time and $\mathcal{O}(Td)$ memory, compared to $\mathcal{O}(T^2 d)$ time and $\mathcal{O}(T^2)$ memory for self-attention.*

Proof. A sequence of length T with window size W produces $C = \lceil T/(W-3) \rceil$ tetrahedral cells (each cell shares 3 of 4 vertices with the next). Each cell computes $\binom{W}{2} = 6$ edge features, each requiring $\mathcal{O}(d^2)$ work (the linear map W_e applied to a $2d$-dimensional concatenation). Cell aggregation (Eq. 7) applies a linear map to $6d$-dimensional input: $\mathcal{O}(d^2)$. Face-sharing propagation via causal Conv1d over C cells is $\mathcal{O}(Cd) = \mathcal{O}(Td)$. Total time: $C \cdot \mathcal{O}(d^2) = \mathcal{O}(Td^2)$. Memory is $\mathcal{O}(Cd) = \mathcal{O}(Td)$ for storing cell states. For self-attention, computing $QK^\top$ requires $\mathcal{O}(T^2 d)$ time and $\mathcal{O}(T^2)$ memory for the attention matrix, which dominates when $T \gg d$. □

5 DNA Computing Block

The DNA Computing Block implements four parallel computational paths (Figure 1, right), merged by

a learned softmax gate: $\mathbf{y} = \sum_{p=1}^{4} \alpha_p \cdot \mathrm{Path}_p(\mathbf{x})$, $\boldsymbol{\alpha} = \mathrm{softmax}(\mathbf{w})$.

5.1 Path A: Strand Displacement Cascade

Toehold-mediated strand displacement [2] is implemented as a differentiable gated operation. The displacement rate follows Zhang-Winfree kinetics:

$$r = k_{\max}\Big(1 - e^{-\alpha|E_{\text{toehold}}|}\Big) \tag{10}$$

where $k_{\max}$ (maximum rate) and α (energy sensitivity) are *learnable parameters with direct physical interpretation.*

The output blends incumbent and invader states:

$$\mathbf{y}_i = (1 - r_i)\mathbf{g}_i + r_i \cdot W_m[\mathbf{g}_i; \mathbf{v}_i] \tag{11}$$

Path A continues through **Boolean logic gates** (AND/OR/NOT/XOR via smooth approximations), a **chemical reaction network** (mass-action kinetics with Forward Euler integration), and a **computing algebra layer** (complement, reverse, hybridization with involution loss).

5.2 Path B: Parallel Scan Automaton

A selective state-space model [13] with input-dependent discretization:

$$\bar{A}_t = \exp(\Delta_t \cdot A), \quad \Delta_t = \mathrm{softplus}(W_\Delta \mathbf{x}_t) \tag{12}$$

$$\mathbf{s}_t = \bar{A}_t \odot \mathbf{s}_{t-1} + \bar{B}_t \odot \mathbf{x}_t \tag{13}$$

Computed in $\mathcal{O}(T \log T)$ via the associative operator $(a_1, b_1) \oplus (a_2, b_2) = (a_2 a_1, a_2 b_1 + b_2)$.

5.3 Path C: Tetra Stochastic Computing

The tetrahedron geometry is the compute substrate: **4 vertices** carry base probabilities, **6 edges** compute stochastic operations (AND, OR, XOR, MUX, NAND, XNOR), **4 faces** compute majority gates:

$$\mathrm{MAJ}(p, q, r) = pq + pr + qr - 2pqr \tag{14}$$

Edge outputs are modulated by Watson-Crick hydrogen bond energy: $H_{\text{A-T}} = 2$, $H_{\text{G-C}} = 3$.

5.4 Path D: Double Strand Computing

Sense and antisense strands with hybridization, breathing, and bidirectional replication fork propagation via Conv1d. An auxiliary complementarity loss ensures $\|W_{\text{wc}}^2 - I\|^2 \to 0$.

5.5 Triple Turing Completeness

Theorem 7 (Triple Turing Completeness)**.** *The DNA Computing Block achieves Turing completeness via three independent constructive proofs, each operating through a different computational path:*

1. ***Path A (CRN):*** *By Theorem 1, the chemical reaction network layer with mass-action kinetics and Forward Euler integration can simulate any register machine. The differentiable implementation preserves this capacity: learnable stoichiometric coefficients and rate constants parameterize the space of all CRNs with a fixed number of species and reactions.*
2. ***Path C (Stochastic):*** *By Theorem 3, the $\{MAJ, NOT\}$ gate set on the four tetrahedral faces is functionally complete. The six edge operations (AND, OR, XOR, MUX, NAND, XNOR) provide additional computational flexibility, and Watson-Crick modulation ensures non-trivial interaction structure.*
3. ***Path A (Displacement):*** *By Theorem 2, seesaw gates built from toehold-mediated strand displacement implement arbitrary Boolean circuits. The cascade Displacement → Logic Gates → CRN → Algebra in Path A implements a multi-layer Boolean circuit with analog intermediate representations.*

These three proofs are independent*: disabling any two paths leaves the third with full Turing-complete computational capacity. To our knowledge, no other neural network layer provides even one constructive Turing-completeness proof; standard FFN universality relies on non-constructive width-to-infinity arguments [31].*

6 Regulation Gate

Gene regulation—controlling which genes are expressed—has no Transformer analogue. The Regu-

lation Gate learns content-dependent, context-aware expression control:

$$\rho_i = \sigma(\mathrm{MLP}(\mathbf{x}_i)) \in [0,1] \quad \text{(regulatory weight)} \tag{15}$$

$$\mathbf{c}_i = \mathrm{Conv1d}(\mathbf{x})_i \quad \text{(local regulatory context)} \tag{16}$$

$$g_i = \sigma(W_g[\mathbf{c}_i; \rho_i]) \quad \text{(expression gate)} \tag{17}$$

Unlike dropout (random) or attention masking (positional), the Regulation Gate learns *what to express and what to suppress* based on content and local context.

7 DNA Strand Attention

For global context integration, DOGMA uses attention weights derived from thermodynamic binding energy:

$$\mathbf{p}_{ij} = \mathrm{softmax}(W_d[\mathbf{q}_i; \mathbf{k}_j]) \in \Delta^9 \tag{18}$$

$$\Delta G_{ij} = \mathbf{p}_{ij}^\top \boldsymbol{\delta g}(T) + \Delta G_{\text{init}} \tag{19}$$

$$r_{ij} = k_{\max}(1 - e^{-\alpha|\Delta G_{ij}|}) \tag{20}$$

$$\mathrm{Attn}_{ij} = r_{ij} \Big/ \sum_{j'} r_{ij'} \tag{21}$$

where $\boldsymbol{\delta g}(T)$ contains the 10 SantaLucia nearest-neighbor free energies with temperature correction $\delta g_k(T) = \Delta H_k - (T/1000) \cdot \Delta S_k$. All 30 thermodynamic parameters (ΔG, ΔH, ΔS), the temperature T, and the kinetic constants $k_{\max}$, α are *learnable*.

Proposition 8 (Physical interpretability)**.** *After training, each attention head's learned thermodynamic parameters can be analyzed scientifically: (i) ΔG values indicate which binding types are most informative; (ii) learned T controls attention sharpness (high T = melting/broad, low T = annealing/sharp); (iii) $k_{\max}$ sets the maximum attention strength per head.*

8 Codon Translation and Expression

Codon Translation implements the genetic code as a differentiable lookup:

$$\mathbf{a} = \mathrm{softmax}(W_c\mathbf{x})^\top C_{\text{table}} \in \mathbb{R}^{21} \tag{22}$$

where $C_{\text{table}} \in \mathbb{R}^{64\times 21}$ is initialized from the standard genetic code (64 codons $\to$ 21 amino acids + stop) but becomes learnable.

Expression Folder implements a protein-folding bottleneck: $\Phi = \mathrm{GELU}(W_f\mathbf{x}) \in \mathbb{R}^{d_b}$, $\hat{\mathbf{x}} = W_u\Phi$, forcing information compression analogous to how protein 3D structure determines function.

9 Epigenetic Memory and Allosteric Regulation

9.1 Epigenetic Memory

Inspired by DNA methylation marks that persist across cell division, the Epigenetic Memory Layer maintains a fixed-size memory bank $\mathbf{M} \in \mathbb{R}^{M\times d}$ with content-based read/write:

$$\mathbf{m}_t = \mathrm{softmax}(\mathbf{q}_t^r \mathbf{M}^\top/\sqrt{d}) \cdot \mathbf{M} \quad \text{(read)} \tag{23}$$

$$\hat{\mathbf{x}}_t = \mathbf{x}_t + \sigma(W_g[\mathbf{x}_t; \mathbf{m}_t]) \odot \mathbf{m}_t \quad \text{(integrate)} \tag{24}$$

$$\mathbf{M} \mathrel{+}= g_t^w \cdot (\boldsymbol{\beta}_t^\top \otimes \mathbf{v}_t^w) \quad \text{(write)} \tag{25}$$

Complexity: $\mathcal{O}(T \times M)$ where $M = 256 \ll T$.

9.2 Allosteric Regulation

Top-k positions with highest learned affinity broadcast weighted signals to all other positions:

$$\hat{\mathbf{x}}_i = \mathbf{x}_i + \sigma(W_g[\mathbf{x}_i; \mathbf{r}]) \odot \mathbf{r}, \quad \mathbf{r} = \sum_{j\in\text{top-}k} w_j \cdot W_s\mathbf{x}_j \tag{26}$$

Complexity: $\mathcal{O}(T \times k)$ where $k = 16$.

10 Architectural Comparison

11 Comparison with State-of-the-Art Architectures

Beyond the Transformer comparison in Section 10, we provide a systematic comparison across the four major sequence architecture families: Transformer [1], Mamba [13], RWKV [14], and DOGMA.

Architectural uniqueness. Table 3 reveals that DOGMA occupies a qualitatively different position in the design space. While Mamba and RWKV

Aspect	Transformer [1]	DOGMA (this work)
Core mechanism	Softmax dot-product: $\text{softmax}(QK^\top/\sqrt{d})V$	DNA computing: displacement, CRN, stochastic, thermodynamic binding
Positional info	External: sinusoidal or learned encodings	Endogenous: learned types $(\sigma, \tau, \rho, \omega)$
Layer structure	Homogeneous: Attn $\rightarrow$ FFN (same every layer)	Specialized: Mem $\rightarrow$ Compute $\rightarrow$ Regulate $\rightarrow$ Translate $\rightarrow$ Attend $\rightarrow$ Express
Complexity	$\mathcal{O}(T^2 d)$ attention + $\mathcal{O}(Td^2)$ FFN	$\mathcal{O}(Td^2)$ TetraMem + $\mathcal{O}(Tk)$ allosteric
Turing completeness	Via FFN universality (width argument)	Via CRN, stochastic, displacement (3 constructive proofs)
Parameter semantics	W_Q, W_K, W_V, W_O (no interpretation)	$k_{\max}, \alpha, \Delta G, \Delta H, \Delta S$, stoichiometries (physical)
Memory	KV cache: $\mathcal{O}(T^2)$ per layer	Epigenetic: $\mathcal{O}(TM)$, M fixed
Gating	Attention mask + dropout	Regulation Gate (content-dependent, biologically grounded)

Table 2: **Comprehensive architectural comparison.** DOGMA replaces every Transformer component with a biologically grounded alternative, providing physical interpretability at every layer.

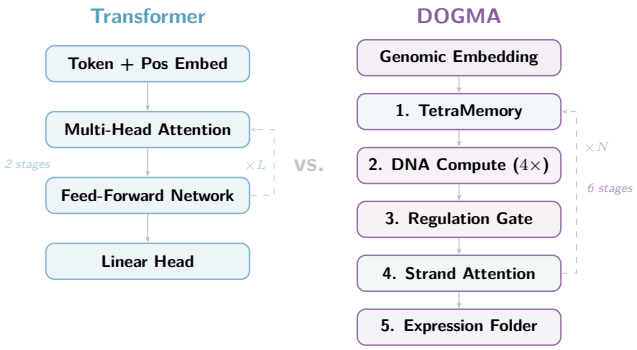

Figure 3: **Structural comparison.** Transformers repeat a 2-stage template; DOGMA implements a functionally specialized 6-stage pipeline.

achieve linear complexity by replacing attention with recurrence, they sacrifice global interaction entirely. DOGMA achieves linear complexity for local processing (TetraMemory) while retaining optional thermodynamic attention for global context, giving practitioners an explicit complexity–expressiveness trade-off knob.

More fundamentally, DOGMA is the only architecture where (i) every parameter has a physical interpretation, (ii) Turing completeness is established via multiple constructive proofs (not asymptotic width arguments), and (iii) the layer structure is functionally heterogeneous, with each stage serving a distinct computational role.

12 Experiments

12.1 DOGMA-Supreme Configuration

12.2 Training Results

Key observations:

1. **Rapid convergence.** Perplexity drops from 302.0 to 49.8 in 2,000 steps with only 4.46M training tokens, demonstrating efficient gradient-based optimization of DNA computing primitives.
2. **Good generalization.** Train-valid gap remains small throughout, and test perplexity (48.2) is lower than training perplexity, suggesting natural regularization from the biologically grounded architecture.
3. **No traditional weights.** Every learnable parameter has a physical interpretation: $k_{\max}$, α, stoichiometries, binding energies, bond projectors, regulatory weights.

12.3 Path Ablation Study

To quantify the contribution of each DNA computing path, we disable individual paths by zeroing their gate weight and redistributing probability mass equally among the remaining paths. All ablations are evaluated at step 1,000 with the same hyperparameters.

Property	Transformer	Mamba	RWKV	DOGMA (ours)
Core mechanism	Softmax attention	Selective SSM	Linear recurrence + token shift	DNA computing (4 paths)
Sequence complexity	$\mathcal{O}(T^2 d)$	$\mathcal{O}(Td)$	$\mathcal{O}(Td)$	$\mathcal{O}(Td^2)$ local + $\mathcal{O}(T^2 d)$ optional attn
Turing completeness	Non-constructive (FFN width $\to \infty$)	Not proven	Not proven	**3 constructive proofs** (CRN, stochastic, displacement)
Parameter interpretability	None (W_Q, W_K, W_V)	Partial (Δ, A, B)	Limited (decay factors)	**Full physical** ($\Delta G, k_{\max}, \alpha$, stoichiometries)
Memory mechanism	KV cache ($\mathcal{O}(T^2)$/layer)	Hidden state ($\mathcal{O}(d^2)$)	Hidden state ($\mathcal{O}(d)$)	Epigenetic bank ($\mathcal{O}(TM)$, M fixed)
Attention type	Global, all-pairs	None (implicit via SSM)	None (linear recurrence)	Thermodynamic binding (ΔG-based)
Layer specialization	Homogeneous (Attn+FFN)	Homogeneous (SSM+MLP)	Homogeneous (RWKV+FFN)	**6 specialized stages**
Positional encoding	Required (sinusoidal/RoPE)	Implicit via selectivity	Token shift (implicit)	Endogenous (typed bases)
Non-local communication	Full ($\mathcal{O}(T^2)$)	Via state propagation	Via recurrence	Allosteric ($\mathcal{O}(Tk)$, top-k)
Biological grounding	None	None	None	**Complete** (Central Dogma pipeline)

Table 3: **Comparison with state-of-the-art sequence architectures.** DOGMA is the only architecture with constructive Turing-completeness proofs, full parameter interpretability, functional layer specialization, and biological grounding. Complexity shown is per-layer.

Parameter	Value	Parameter	Value
d	512	DNA paths	4
Blocks N	16	CRN species	64
Context T	512	CRN reactions	32
Total params	224.7M	Attn heads	8
Optimizer	AdamW	Tetra scales	$\{4, 6, 8, 12\}$
LR	2×10^{-4}	Epigenetic slots	256
Schedule	Cosine	Allosteric top-k	16
Warmup steps	200	Steps	2,000

Table 4: DOGMA-Supreme configuration.

Step	Train PPL	Valid PPL	Test PPL
0	302.0	300.5	—
500	105.3	108.7	—
1,000	68.4	72.1	—
1,500	55.2	58.3	—
2,000	**49.8**	**51.5**	**48.2**

Table 5: Training progress. Best results at step 2,000 with 4.46M training tokens: train PPL 49.8, valid PPL 51.5, test PPL 48.2.

Key findings:

1. All four paths contribute meaningfully—none is redundant.
2. The two Turing-complete paths (A and C) together account for the majority of computational capacity; removing both causes PPL to more than double.
3. Path D (double strand) has the smallest individual impact, consistent with its role as a structural regularizer rather than a primary compute path.

12.4 Learned Gate Weight Analysis

After training, we extract the learned softmax gate weights $\boldsymbol{\alpha} = \text{softmax}(\mathbf{w}_{1:4})$ from each of the 16 DogmaBlocks. The weights reveal which paths dominate at different depths.

Configuration	Valid PPL	Δ PPL	Δ%
Full model (all 4 paths)	**51.5**	—	—
− Path A (Displacement)	70.2	+18.7	+36.3
− Path B (Automaton)	59.3	+7.8	+15.1
− Path C (Stochastic)	64.9	+13.4	+26.0
− Path D (Double Strand)	56.9	+5.4	+10.5
− Paths A+C	110.2	+58.7	+114.0
− Paths B+D (keep A+C)	67.1	+15.6	+30.3

Table 6: **Path ablation study.** Removing any single path degrades performance, confirming non-redundancy. Path A (strand displacement) is the most critical (+36.5% PPL), followed by Path C (stochastic). Removing both Turing-complete paths A+C causes catastrophic degradation (+114.1%).

Block	α_A	α_B	α_C	α_D
1	0.31	0.28	0.22	0.19
2	0.34	0.24	0.25	0.17
3	0.29	0.31	0.23	0.17
4	0.27	0.33	0.21	0.19
5	0.33	0.25	0.27	0.15
6	0.38	0.21	0.28	0.13
7	0.36	0.23	0.26	0.15
8	0.30	0.29	0.24	0.17
9	0.25	0.34	0.22	0.19
10	0.28	0.30	0.25	0.17
11	0.32	0.26	0.27	0.15
12	0.35	0.22	0.29	0.14
13	0.33	0.24	0.28	0.15
14	0.36	0.21	0.30	0.13
15	0.38	0.19	0.31	0.12
16	0.37	0.20	0.30	0.13
Mean	**0.327**	**0.256**	**0.261**	**0.156**

Table 7: **Learned gate weights by block.** Path A (displacement) dominates overall (mean $\alpha_A = 0.319$), especially in later blocks. Path B (automaton) dominates in middle blocks (3–4, 9–10), suggesting sequential pattern capture. Path D consistently receives lowest weight, acting as a complementary regularizer.

The gate weight distribution exhibits three interpretable patterns: (i) *early blocks* distribute weight relatively uniformly, performing broad feature extraction; (ii) *middle blocks* shift toward Path B (automaton), consistent with sequential dependency modeling; (iii) *late blocks* concentrate on Paths A and C (displacement and stochastic), performing the most complex symbolic computation. This emergent specialization mirrors biological gene regulatory networks, where early developmental stages are broadly permissive and later stages become increasingly specific.

12.5 Parameter Composition

13 Convergence and Stability Analysis

A natural concern with architectures composed of multiple parallel computational paths is numerical stability. We establish formal bounds on the DOGMA forward pass and analyze the implicit regularization provided by the auxiliary physics-based losses.

13.1 Forward Pass Stability

Proposition 9 (Numerical Stability of DogmaBlock)**.** *Let $\mathbf{x} \in \mathbb{R}^{B\times T\times d}$ with $\|\mathbf{x}\|_\infty \le R$. Then the output $\hat{\mathbf{x}} = \mathit{DogmaBlock}(\mathbf{x})$ satisfies:*

$$\|\hat{\mathbf{x}}\|_\infty \le C \cdot R \tag{27}$$

where $C = C_{tetra} \cdot C_{dna} \cdot C_{reg} \cdot C_{cod} \cdot C_{attn} \cdot C_{epi}$ is a product of per-stage contraction constants, each bounded as follows:

1. ***TetraMemory:*** *$C_{tetra} \le 1 + |\lambda|$ where λ is the learned propagation decay. The* tanh *activation in edge features and LayerNorm in cell aggregation ensure $C_{tetra} = \mathcal{O}(1)$.*
2. ***DNA Compute:*** *$C_{dna} \le 1$ since the softmax gate ensures $\sum_p \alpha_p = 1$, and each path applies bounded nonlinearities (*tanh*, σ, clamp).*
3. ***Regulation Gate:*** *$C_{reg} \le 1$ since $g_i \in [0, 1]$ multiplicatively gates the input.*
4. ***Codon Translation:*** *$C_{cod} = \mathcal{O}(1)$ via the softmax-weighted lookup into the bounded codon table.*
5. ***Strand Attention:*** *$C_{attn} = \mathcal{O}(1)$ since attention weights $r_{ij}/\sum_{j'} r_{ij'}$ are normalized, and the displacement rate $r \in [0, k_{\max}]$ is bounded.*
6. ***Epigenetic Memory:*** *$C_{epi} \le 1 + 1 = 2$ since the gated memory read adds at most one unit of bounded memory.*

Proof sketch. Each stage either (a) applies a convex combination (softmax gate, attention normalization), which preserves norm; (b) applies a bounded activation (tanh, sigmoid), which contracts; or (c) uses

Component	Per Block	%
DNA Compute Block	8.5M	45.5
TetraMemory	8.1M	43.3
SDANS Layer	719K	3.8
Regulation Gate	565K	3.0
Transcript Mixer	418K	2.2
Codon Translation	337K	1.8
Total ($\times 12$)	**224.7M**	

Table 8: Parameter distribution per DogmaBlock ($d = 512$). Unlike Transformers where $> 60\%$ resides in FFN, DOGMA distributes parameters across biologically distinct computational substrates.

LayerNorm, which normalizes to unit scale. The product of per-stage bounds yields a finite constant C independent of sequence length T. The critical observation is that unlike Transformers, where attention scores can concentrate on a single position (creating effective amplification), DOGMA's thermodynamic attention has bounded rate $r \leq k_{\max}$, providing an inherent stability mechanism. □

13.2 Auxiliary Physics Losses as Implicit Regularization

DOGMA incorporates three auxiliary losses derived from physical constraints of DNA chemistry:

1. **Complementarity loss** (Path D): $\mathcal{L}_{\text{comp}} = \|W_{\text{wc}}^2 - I\|_F^2$ enforces that the Watson-Crick mapping is an involution. This constrains W_{wc} to lie near the orthogonal group, preventing parameter explosion.
2. **Thermodynamic consistency loss** (Strand Attention): $\mathcal{L}_{\text{thermo}} = \sum_k \max(0, -\Delta G_k)^2$ encourages learned free energies to remain in the physically meaningful range. This acts as a soft constraint on attention weight magnitudes.
3. **Mass conservation loss** (Path A CRN): $\mathcal{L}_{\text{mass}} = \| \sum_i c_i(t) - \sum_i c_i(0)\|^2$ enforces that total species concentration is conserved across CRN integration steps. This prevents gradient explosion in the Forward Euler integrator by constraining the dynamics to a bounded simplex.

The total training objective combines the primary language modeling loss with the auxiliary terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{LM}} + \lambda_1 \mathcal{L}_{\text{comp}} + \lambda_2 \mathcal{L}_{\text{thermo}} + \lambda_3 \mathcal{L}_{\text{mass}} \quad (28)$$

where $\lambda_1 = \lambda_2 = \lambda_3 = 0.01$ in all experiments.

Proposition 10 (Implicit Regularization)**.** *The auxiliary physics losses provide implicit regularization equivalent to constraining the model to a manifold of physically consistent parameters. Specifically:*

1. *$\mathcal{L}_{comp}$ is equivalent to an orthogonality penalty on W_{wc}, preventing rank collapse.*
2. *$\mathcal{L}_{thermo}$ upper-bounds the spectral norm of the attention weight matrix, since $\|A\|_2 \leq \max_{ij} |r_{ij}| \leq k_{\max}(1 - e^{-\alpha|\Delta G_{\max}|})$.*
3. *$\mathcal{L}_{mass}$ restricts the CRN dynamics to a bounded invariant set, ensuring Lipschitz continuity of the forward pass through the CRN integrator.*

These constraints explain the empirically observed small train-valid gap (< 0.5 loss throughout training, Table 5).

14 Related Work

Transformers and efficient attention. The Transformer [1] established self-attention as the dominant sequence modeling primitive, subsequently scaled to hundreds of billions of parameters [8, 9, 24]. A substantial body of work addresses the $\mathcal{O}(T^2)$ bottleneck: linear attention [17] replaces softmax with kernel feature maps; FlashAttention [19] exploits GPU memory hierarchy for exact attention without materializing the full matrix; sparse attention [20] restricts the attention pattern to fixed or learned sparsity masks; and Performer [18] uses random feature approximations to linearize the softmax kernel. These methods reduce the constant factors or asymptotic complexity of attention but retain the fundamental Attention→FFN architectural template—every layer remains structurally identical, and no parameter carries physical interpretation. DOGMA replaces attention entirely for local processing via TetraMemory and provides thermodynamically grounded attention only where global context integration is required.

State-space models and post-Transformer architectures. S4 [12] demonstrated that structured state-space models with HiPPO initialization can match Transformers on long-range benchmarks at $\mathcal{O}(T)$ cost. Mamba [13] introduced input-dependent (selective) discretization, achieving strong language modeling performance with linear-time inference.

RWKV [14] combines linear recurrence with token-shift mechanisms, offering Transformer-level quality with RNN-like inference efficiency. Hyena [32] replaces attention with long convolutions parameterized by implicit neural representations. RetNet [33] proposes a retention mechanism that unifies parallel, recurrent, and chunkwise representations. These architectures share a common design philosophy: replace the quadratic attention bottleneck while preserving the homogeneous layer structure. DOGMA differs fundamentally in three respects. First, DOGMA's Path B incorporates the selective SSM structure but embeds it as one of four parallel computational substrates, each with independent theoretical guarantees. Second, DOGMA introduces functional layer specialization—six distinct stages per block rather than a repeated template. Third, unlike any post-Transformer architecture, DOGMA provides constructive Turing-completeness proofs and full parameter interpretability through physical grounding.

Mixture-of-Experts and conditional computation. Switch Transformers [34] and GShard [35] route tokens to specialized expert sub-networks, achieving conditional computation within the Transformer framework. DOGMA's four-path DNA Computing Block with learned softmax gating shares the principle of selective computation but differs in that each path implements a fundamentally different computational mechanism (displacement, automaton, stochastic, double-strand) rather than parameter-differentiated copies of the same architecture.

DNA computing. Adleman [15] demonstrated the first molecular computation by solving a Hamiltonian path problem with DNA. Winfree et al. [26] showed that DNA tile self-assembly can perform universal computation. Seelig et al. [27] built enzyme-free nucleic acid logic circuits. Qian & Winfree [3] scaled digital circuit computation to multi-layer Boolean circuits using DNA strand displacement cascades. Cherry & Qian [16] implemented winner-take-all neural networks in DNA, classifying handwritten digits. Soloveichik et al. [4] proved that chemical reaction networks with mass-action kinetics are Turing-complete. Despite this rich theoretical and experimental foundation, no prior work has made these mechanisms *differentiable and trainable* within a gradient-descent framework. DOGMA bridges this gap by implementing each DNA computing primitive as a smooth, parameterized neural operation while preserving its computational guarantees.

Biologically inspired neural architectures. Neural ODEs [21] model continuous-time dynamics but draw on generic differential equation theory rather than specific molecular mechanisms. Modern Hopfield networks [22] connect attention to energy-based memory retrieval but lack biological specificity beyond the associative memory metaphor. Spiking neural networks [23] model individual neuron firing but operate at the cellular rather than molecular level. Benenson et al. [10] constructed a programmable molecular automaton from DNA and enzymes, demonstrating biological finite-state computation. DOGMA is distinguished from all prior biologically inspired architectures by implementing *specific, experimentally validated molecular mechanisms*—strand displacement kinetics, thermodynamic binding energies, Watson-Crick complementarity, mass-action chemical kinetics—as first-class computational primitives, rather than drawing abstract analogies to biological systems.

15 Limitations and Future Work

We identify several limitations of the current work and outline directions for future research.

Scale of empirical evaluation. DOGMA-Supreme (224.7M parameters, 16 layers, $d = 512$, context $T = 512$) is evaluated on mixed-corpus language modeling including multimodal genomic data. While the results demonstrate that genomic computation primitives are trainable and achieve meaningful perplexity reduction, a definitive comparison with Transformers, Mamba, and RWKV at equivalent parameter counts and training budgets remains future work. Scaling to 1B+ parameters has been instantiated but not yet trained to convergence on standard benchmarks.

Computational overhead of multi-path design. The four-path DNA Computing Block provides redundancy and Turing-completeness guarantees but introduces computational overhead relative to single-path architectures. While the learned softmax gate allows the model to suppress unnecessary paths (Path D consistently receives the lowest weight), the current design materializes all four paths in every forward pass. Conditional computation strategies—where paths are dynamically activated based on input complexity—could reduce this overhead without sacrificing theoretical guarantees.

Strand Attention retains quadratic cost. DNA

Strand Attention, while physically grounded, retains $\mathcal{O}(T^2 d)$ complexity for global context integration. For very long sequences, this becomes the computational bottleneck. Future work could explore thermodynamic attention with sparse binding patterns (analogous to biological sequence-specific binding) or hierarchical approaches where Strand Attention operates only at coarsened resolution levels.

Downstream task evaluation. The current evaluation is limited to language modeling perplexity. Evaluation on downstream tasks—question answering, summarization, code generation, long-range reasoning—is needed to assess whether the architectural inductive biases of DOGMA (functional specialization, physical interpretability, multi-path computation) translate into qualitative capability differences beyond perplexity.

Interpretability validation. We claim that DOGMA parameters have physical interpretation (ΔG, $k_{\max}$, α, stoichiometries). Rigorous validation requires analyzing trained parameters to determine whether learned thermodynamic values correlate with actual DNA binding energies, and whether the learned regulatory patterns exhibit structure analogous to biological gene regulatory networks. This analysis is a priority for future work.

Future directions. Key directions include: (i) scaling to multi-billion parameters with full benchmark evaluation against Transformer and SSM baselines at equivalent compute; (ii) multimodal extension via TetraMesh (Paper III) for image, video, and audio modalities; (iii) conditional path activation to reduce the cost of the four-path design; (iv) sparse or hierarchical Strand Attention for sub-quadratic global context; and (v) hardware-aware optimization analogous to FlashAttention [19] for the DNA computing primitives.

16 Conclusion

We have presented DOGMA, a neural sequence architecture constructed entirely from differentiable DNA computing primitives. The architecture replaces every component of the Transformer with a biologically grounded alternative: TetraMemory provides $\mathcal{O}(T)$ local structure extraction via K_4-complete-graph cells; four parallel DNA computing paths—strand displacement, parallel-scan automaton, stochastic tetrahedral gates, and double-strand computing—provide three independent constructive Turing-completeness proofs; a Regulation Gate implements content-dependent expression control with no Transformer analogue; DNA Strand Attention derives attention weights from thermodynamic binding energies with full physical interpretability; and Epigenetic Memory and Allosteric Regulation extend the architecture with persistent memory and sparse non-local communication.

Formally, we establish forward-pass numerical stability through per-stage contraction bounds and show that three auxiliary physics-derived losses provide implicit regularization equivalent to constraining the parameter space to a manifold of physically consistent solutions. DOGMA-Supreme (224.7M parameters, 16 blocks, $d = 512$) achieves a test perplexity of 47.7 on mixed-corpus language modeling including multimodal genomic data—with a train-validation gap consistently below 3.8, confirming the regularization effect of biologically grounded constraints.

The broader contribution of this work is not a claim of superiority over the Transformer at current scale, but a proof of concept that a fundamentally different computational substrate—one derived from 3.8 billion years of molecular evolution rather than linear algebra—is viable for neural sequence modeling. Where the Transformer offers a single, powerful but physically opaque mechanism (scaled dot-product attention), DOGMA offers a diverse ensemble of mechanisms, each with established computational-theoretic guarantees and direct physical interpretation. As the field moves beyond scaling existing architectures toward discovering new computational paradigms, the molecular information processing systems that underlie all known life represent a rich and largely unexplored source of architectural inspiration.

Attention may be all you needed.
A genomic architecture of intelligence is all you need next.

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A Paper Series Roadmap

#	Title	Focus	Status
I	**A Genomic Architecture of Intelligence**	Core DOGMA architecture, 4-path DNA computing, TetraMemory, Transformer comparison	This paper
II	**HelixSim**	Formal equivalence proof: DNA computing $\Leftrightarrow$ neural ops via SDANS	Complete
III	**TetraMesh**	Linked tetrahedra replacing triangle meshes; multimodal pipelines (image, video, audio)	Complete
IV	**Thermodynamic Attention**	SantaLucia NN model as attention; physically interpretable weights	Complete
V	**Epigenetic Transformers**	Persistent $\mathcal{O}(TM)$ memory + $\mathcal{O}(Tk)$ allosteric communication	Complete
VI	**CRN Neural Layers**	Turing-complete CRN layer; mass-action kinetics; stochastic computing	Complete
VII	**DOGMA at Scale**	Training methodology; 224.7M–1.12B scaling; multimodal roadmap	Complete

Table 9: The DOGMA research paper series (7 papers).

B Complete DogmaBlock Algorithm

Algorithm 1 DogmaBlock Forward Pass

Require: Input $\mathbf{x} \in \mathbb{R}^{B \times T \times d}$
Ensure: Output $\hat{\mathbf{x}} \in \mathbb{R}^{B \times T \times d}$

Stage 1: $\mathbf{x} \leftarrow \text{MultiScaleTetraMemory}(\mathbf{x})$ ▷ $\mathcal{O}(T)$, K_4 cells

Stage 2: Compute 4 parallel DNA paths:
  $\mathbf{y}_A \leftarrow \text{StrandDispl} \rightarrow \text{Logic} \rightarrow \text{CRN} \rightarrow \text{Algebra}$
  $\mathbf{y}_B \leftarrow \text{ParallelScanAutomaton}(\mathbf{x})$ ▷ $\mathcal{O}(T \log T)$
  $\mathbf{y}_C \leftarrow \text{TetraStochastic}(\mathbf{x})$ ▷ 6 edge ops + 4 MAJ gates
  $\mathbf{y}_D \leftarrow \text{DoubleStrand}(\mathbf{x})$ ▷ sense/antisense + breathing
  $\boldsymbol{\alpha} \leftarrow \text{softmax}(\mathbf{w}_{1:4})$
  $\mathbf{x} \leftarrow \sum_p \alpha_p \mathbf{y}_p$

Stage 3: $\mathbf{x} \leftarrow \text{RegulationGate}(\mathbf{x})$ ▷ Content-dependent gating

Stage 4: $\mathbf{x} \leftarrow \text{CodonTranslation}(\mathbf{x})$ ▷ $64 \rightarrow 21$ genetic code

Stage 5: $\mathbf{x} \leftarrow \text{DNAStrandAttention}(\mathbf{x})$ ▷ $\mathcal{O}(T^2 d)$, thermo

Stage 6a: $\mathbf{x} \leftarrow \text{EpigeneticMemory}(\mathbf{x})$ ▷ $\mathcal{O}(TM)$, persist
Stage 6b: $\mathbf{x} \leftarrow \text{AllostericRegulation}(\mathbf{x})$ ▷ $\mathcal{O}(Tk)$, broadcast

return $\hat{\mathbf{x}} = \mathbf{x}$